Subject Description Form

Subject Code	DSAI4209
Subject Title	Introduction to Generative AI and Foundation Models
Credit Value	3
Level	4
Pre-requisite/ Co-requisite/ Exclusion	Knowledge in machine learning fundamentals, Python programming, college linear algebra, probability theory, and basic deep learning concepts
Objectives	The objectives of this course are to provide a comprehensive introduction to the principles, architectures, and applications of foundation models and generative AI. The course aims to equip students with both theoretical understanding and practical skills to develop, adapt, and deploy large-scale generative models across language, vision, and multimodal domains, while critically evaluating their capabilities, limitations, and societal implications.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. Explain the core principles, architectures, and theoretical foundations of generative models and foundation models, including transformers, diffusion models, and state-space models. b. Analyze the scaling laws, emergent abilities, and theoretical properties of large-scale models, and assess statistical and computational trade-offs in model design. c. Implement, fine-tune, and deploy foundation models using modern techniques including parameter-efficient fine-tuning (LoRA, QLoRA), reinforcement learning from human feedback (RLHF), and retrieval-augmented generation (RAG). d. Design and develop applications leveraging generative AI for text, image, code, and multimodal generation, while critically evaluating model performance, biases, and ethical considerations.
Subject Synopsis/ Indicative Syllabus	**Part I: Foundations of Generative Modelling** **Introduction to Generative AI** • Taxonomy of generative models • Foundation models overview • Statistical learning theory basics **Introduction to Classical Models** • Sequence modelling • Encoder-decoder architectures • Transformer Architectures • Variational Autoencoders • Diffusion Models • Generative Adversarial Networks

	<u>Part II: Foundation Models and Scaling</u> **Large Language Models** • BERT, GPT, T5 architectures; • Pretraining objectives; • Scaling laws and emergent abilities **Prompting and In-Context Learning** • Zero-shot and few-shot prompting • Retrieval-Augmented Generation • Chain-of-Thought • Instruction following • Tool integration **Fine-Tuning and Adaptation** • Distributed training • Supervised fine-tuning; • Parameter-efficient fine-tuning (LoRA, QLoRA, adapters) • Reinforcement Learning from Human Feedback (RLHF) **<u>Part III: Advanced Topics and Applications</u>** **Multimodal Foundation Models and Agentic AI** • Vision transformers (ViT) • CLIP; BLIP, LLaVA • Image-text joint embedding • World models • Autonomous agents **State-Space Models and Mamba** • Linear state-space models • Structured state-space models • Mamba architecture **Safety, Ethics, and Societal Impact** • Bias and fairness • Hallucinations • Adversarial attacks • Privacy • Regulatory considerations • Evaluation and Benchmarking
Teaching/Learning Methodology	The subject will be delivered through lectures, tutorials, and hands-on coding labs. Lectures introduce theoretical foundations and architectural principles, while tutorials provide guided problem-solving and paper discussions. Coding labs offer practical experience with implementing algorithms and applying state-of-the-art models using PyTorch and Hugging Face libraries. The teaching approach is problem-oriented, emphasizing both theoretical understanding and practical application. Students are encouraged to engage critically with primary research literature and develop innovative solutions to real-world problems using generative AI. Group discussions and project-based learning foster collaborative problem-solving and peer learning.

Assessment Methods in Alignment with Intended Learning Outcomes

Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)			
		a	b	c	d
1. Assignments	20%	√	√	√	
2. Course Project	40%		√	√	√
3. Quizzes	20%	√	√		
4. Paper Review and Presentation	20%	√			√
Total	100 %				

Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:

Assignments: Assessment of the understanding of core concepts and the ability to implement fundamental algorithms. Assignments include programming tasks implementing generative models from scratch and applying existing frameworks to solve specific problems.

Course Project: Assessment of the ability to develop novel methods or applications using generative AI for real-world problems. Students work in groups to design, implement, and evaluate a complete system, presenting their results in written reports and oral presentations. This assesses synthesis of knowledge, practical implementation skills, and critical evaluation abilities.

Quizzes: Assessment of the comprehension of fundamental concepts, principles, algorithms, and theories through closed-book or open-book questions targeting theoretical understanding and analytical skills.

Paper Review and Presentation: Assessment of the ability to critically read and evaluate primary research literature, communicate complex ideas effectively, and engage with emerging trends in the field.

Student Study Effort Expected

Class contact:	
▪ Lecture	26Hrs.
▪ Tutorial / Lab	13Hrs.
Other student study effort:	
▪ Assignments and Project Work	40Hrs.
▪ Self-study and Reading	51Hrs.
Total student study effort	130Hrs.

Reading List and References

Bommasani, R., Hudson, D. A., Adeli, E., Altman, R., Arora, S., von Arx, S., ... & Liang, P. (2021). **On the opportunities and risks of foundation models**. *arXiv preprint arXiv:2108.07258*.

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intelligence: a systematic review and applications. *Multimedia tools and applications*, *84*(21), 23661-23700.

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., ... & Polosukhin, I. (2017). **Attention is all you need.** In *Advances in neural information processing systems*, *30*, 5998-6008.

Ho, J., Jain, A., & Abbeel, P. (2020). **Denoising diffusion probabilistic models.** In *Advances in neural information processing systems*, *33*, 6840-6851.

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Brown, T., Mann, B., Ryder, N., Subbiah, M., Kaplan, J. D., Dhariwal, P., ... & Amodei, D. (2020). **Language models are few-shot learners.** In *Advances in neural information processing systems*, *33*, 1877-1901.

Kaplan, J., McCandlish, S., Henighan, T., Brown, T. B., Chess, B., Child, R., ... & Amodei, D. (2020). **Scaling laws for neural language models.** *arXiv preprint arXiv:2001.08361*.

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Ouyang, L., Wu, J., Jiang, X., Almeida, D., Wainwright, C., Mishkin, P., ... & Lowe, R. (2022). **Training language models to follow instructions with human feedback.** In *Advances in neural information processing systems*, *35*, 27730-27744.

Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., ... & Houlsby, N. (2021). **An image is worth 16x16 words: Transformers for image recognition at scale.** In *International conference on learning representations*.

Park, J. S., O'Brien, J., Cai, C. J., Morris, M. R., Liang, P., & Bernstein, M. S. (2023). **Generative agents: Interactive simulacra of human behavior.** In *Proceedings of the 36th annual ACM symposium on user interface software and technology* (pp. 1-22).

Gu, A., Goel, K., & Ré, C. (2021). **Efficiently modeling long sequences with structured state spaces.** In *International conference on learning representations*.

Gu, A., & Dao, T. (2024). **Mamba: Linear-time sequence modeling with selective state spaces.** In *First conference on language modeling*.

Bender, E. M., Gebru, T., McMillan-Major, A., & Shmitchell, S. (2021). **On the dangers of stochastic parrots: Can language models be too big?**. In *Proceedings of the 2021 ACM conference on fairness, accountability, and transparency* (pp. 610-623).

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